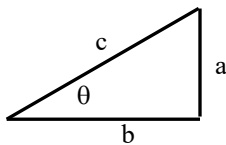


If this were a quiz, it would be closed book, and closed notes. Calculators are permitted only to do mathematical calculations. The **ONLY** physics formulas that may be used are those given below. Define all symbols and justify all mathematical expressions used. Make sure to state all assumptions & approximations used to solve a problem. Partial credit would be given for a well-communicated problem-solving framework based on correct physics. On a quiz, each problem is worth 25 points: **In the context of a unified solution**, credit will be awarded as follows: a useful sketch that shows all relevant quantities, listing the known quantities, and writing the quantity to be determined is 7 points; writing the physics principles useful to solve the problem, writing any assumptions or approximations needed, drawing useful physics diagrams, and writing the equations that apply the physics principles in terms of the quantities from your sketch is 6 points; planning the solution by describing how to use those physics equations to construct an algebraic answer is 5 points; solving those equations for a symbolic result is 3 points; checking the units & reasonableness of the symbolic result is 2 points; calculating a numerical result & stating why its units are correct and giving a reason why that result is not unreasonable is 2 points. If no numerical answer is required, 1 point will be added for the planning the solution and 1 point to solving the equations and checking the units & reasonableness of the solution.

Useful Mathematical Relationships:



For a right triangle: $\sin \theta = \frac{a}{c}$, $\cos \theta = \frac{b}{c}$, $\tan \theta = \frac{a}{b}$,

$$a^2 + b^2 = c^2, \sin^2 \theta + \cos^2 \theta = 1$$

For a circle: $C = 2\pi R$, $A = \pi R^2$

For a sphere: $V = \frac{4}{3}\pi R^3$, $S = 4\pi R^2$

If $Ax^2 + Bx + C = 0$, $x = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$, $(a + b)^2 = a^2 + 2ab + b^2$

Fundamental Concepts, Principles, and Definitions:

$\sum \vec{F} = m\vec{a}$	$\vec{F}_{A \text{ on } B} = -\vec{F}_{B \text{ on } A}$	$a_{x \text{ average}} = \frac{\Delta v_x}{\Delta t}$	$v_{x \text{ average}} = \frac{\Delta x}{\Delta t}$
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Under Certain Conditions:

$F = mg$	$F = \mu N$	$F_{\max} = \mu N$	$F = k\Delta L$
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Useful constants: 1 mile = 5280 ft, $g = 9.8 \text{ m/s}^2 = 32 \text{ ft/s}^2$

Your team has been asked to review the safety of procedures at a construction site that might contribute to excessive back injuries that have been reported. You notice that during one operation, a construction worker guides 3-foot-tall heavy crates down a concrete ramp. To make sure the crate does not run away after it starts sliding down the ramp, the worker stays in front of it and pushes so it moves down the ramp at a constant velocity. To check the safety of this process, you decide to calculate the force that the worker must exert on the crate. For each operation you look up the weight of the crate, the angle of the ramp from the horizontal, and both the coefficient of kinetic and static friction between the bottom of the crate and the ramp. You also measure the height of each worker, their weight, and the angle to the ramp that they must push on the crate.